

Climate 5000 Series - Gen 4

12K BTU Single Zone Regular System

1-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

12K Btu Single Zone Regular System - 1-Way Cassette IDU

Indoor part number	8-733-962-667
Indoor model number	BMS500-AAU012-1AHZXD
Outdoor part number	8-733-962-685
Outdoor model number	BMS500-AAS012-1CSXRD

Efficiency Data		AHRI 210/240 2023
SEER2		22.1
EER2		12.3
HSPF2 Region IV		10
HSPF2 Region V		7.8
COP(47F)		3.40
COP (5F)		2.48

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3900-13800
	Rated Capacity	12000
	Current	4.4
	Input Power	976
Heating	Capacity Range	4700-15300
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	8600
	Current	4.7
	Input Power	1034

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 12.0
Max. fuse indoor/outdoor (A)	15.0 / 15.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 310
Input (W)	325 / 826
Rated current (RLA/A)	2.40 / 5.65

Piping	
Refrigerant precharge (oz/kg)	32.45 / 0.92
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82.00 / 25
Max. difference in height (ft/m)	49.20 / 15

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		3.25
IDU	Fan motor model	ZKFN-30-8-43
	Fan motor qty (ea)	1
	Fan motor input (W)	46.0
	Fan speed (RPM)	1184 / 1064 / 944
	Air flow, Turbo / H / M / L / Si (CFM)	353.2 / 312.0 / 282.5 / 247.2 / 176.6
	Approx. Air Throw Range ft (m)	23.0 / 7
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 43.0 / 37.5 / 26.0 / 22.0
	Net weight (lbs / kg)	45.19 / 20.5
ODU	Gross weight (lbs / kg)	82.45 / 37.4
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-34-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	800 / 700 / 600
	Air flow, Max (CFM)	1236.1
	Noise level, Max (dB(A))	54.5
	Net weight (lbs / kg)	62.17 / 28.2
	Gross weight (lbs / kg)	67.68 / 30.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Standard Parts

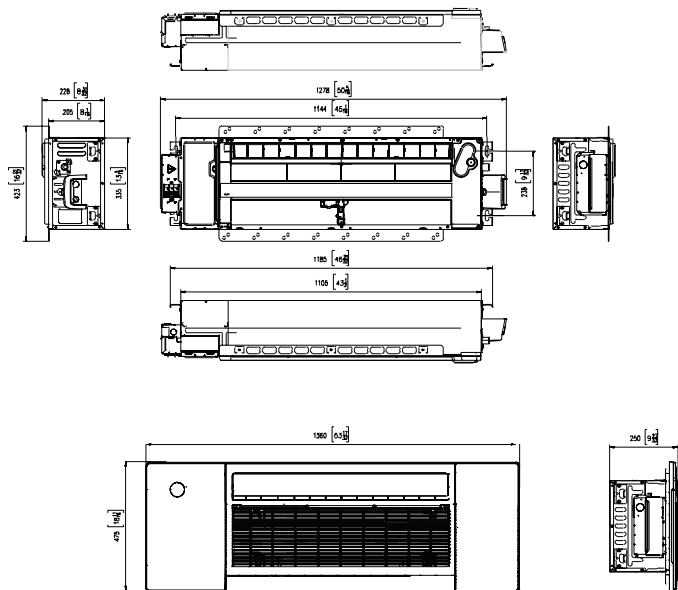
8-733-969-065	Remote Control
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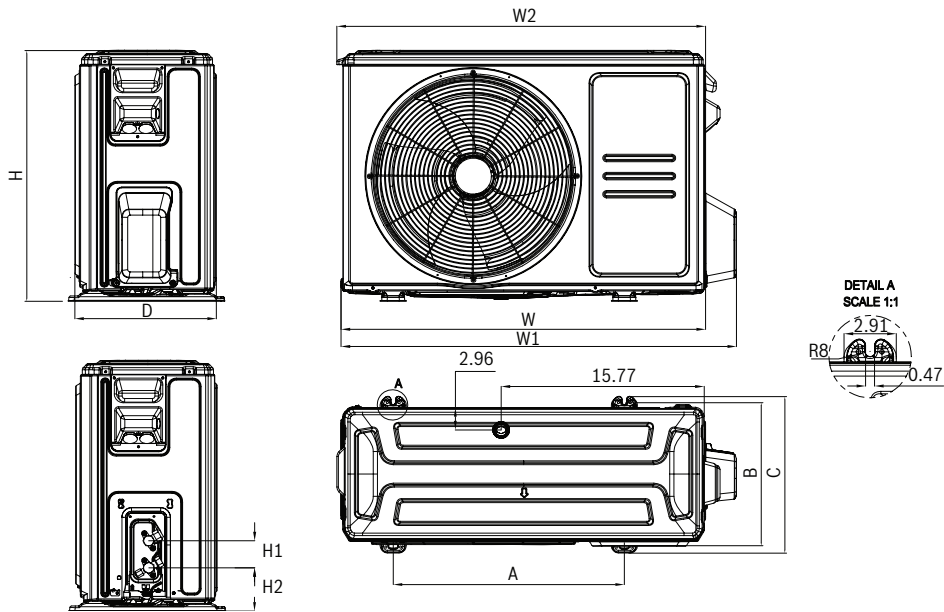


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU012-1AHZXD	50.31	13.19	8.98

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS012-1CSXRD	30.12	11.9	21.85	32.9	30.87	17.8	11.26	12.36	2.36	3.67

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Cooling Performance - Indoor Dry Bulb Temperature at 353 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	S/T	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
-13°F (-25.0°C)	TC	11640	11634	11800	11849	12175	12144	12134	12127	12126	12116	12508	12492	12480	12478	12466	13193	13189	13162	13157	13130
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58
-4°F (-20.0°C)	TC	12535	12529	12708	12761	13112	13079	13068	13060	13058	13048	13470	13453	13440	13438	13425	14208	14204	14175	14170	14140
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.61	0.61	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.61	0.60	0.61	0.61	0.61
0°F (-17.8°C)	TC	12951	12944	13184	13239	13602	13507	13492	13481	13480	13615	13904	13884	13867	13865	13858	14651	14648	14647	14649	14658
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.62
5°F (-15.0°C)	TC	13323	13368	13674	13737	14160	13947	13951	13952	13966	14174	14393	14397	14400	14400	14420	15238	15233	15241	15243	15249
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.64	0.64	0.64	0.64	0.64
14°F (-10.0°C)	TC	14291	14284	14575	14643	15097	15000	14987	14977	14975	15115	15514	15495	15479	15476	15461	16496	16491	16460	16454	16419
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
17°F (-8.3°C)	TC	14578	14626	14964	15037	15525	15304	15310	15313	15326	15542	15826	15832	15839	15839	15859	16832	16827	16824	16828	16842
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.97	0.97	0.97	0.97	0.97
23°F (-5.0°C)	TC	13941	14001	14364	14435	14910	14647	14652	14653	14673	14926	15153	15158	15162	15162	15187	16112	16107	16118	16121	16130
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.88	0.88	0.88	0.88	0.89	0.88	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.91	0.91	0.91	0.91	0.91
35°F (1.7°C)	TC	12792	12792	12867	12903	13319	13389	13389	13390	13390	13426	13809	13808	13808	13809	13810	14586	14583	14582	14581	14583
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.68	0.68	0.68	0.68	0.71	0.72	0.72	0.72	0.72	0.72	0.75	0.75	0.75	0.75	0.75	0.81	0.81	0.81	0.81	0.81
41°F (5.0°C)	TC	12396	12396	12495	12582	12953	12972	12972	12973	12973	13018	13377	13376	13377	13377	13378	14128	14124	14123	14123	14124
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.54	0.54	0.55	0.55	0.58	0.58	0.58	0.58	0.58	0.58	0.60	0.60	0.60	0.60	0.60	0.66	0.66	0.66	0.66	0.66
47°F (8.3°C)	TC	11999	12007	12168	12223	12584	12556	12555	12556	12555	12614	12946	12944	12944	12945	12945	13670	13667	13666	13665	13667
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.41	0.41	0.42	0.42	0.44	0.44	0.44	0.44	0.44	0.44	0.46	0.46	0.46	0.46	0.46	0.50	0.50	0.50	0.50	0.50
50°F (10.0°C)	TC	11699	12021	12427	12499	12983	12359	12366	12452	12491	12996	12842	12858	12865	12866	12990	13727	13763	13798	13803	13820
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47
59°F (15.0°C)	TC	11443	11845	12265	12340	12842	12107	12124	12262	12353	12855	12607	12620	12624	12642	12865	13537	13564	13591	13594	13606
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51
68°F (20.0°C)	TC	11184	11661	12093	12170	12687	11840	11866	12107	12184	12699	12352	12361	12363	12394	12710	13309	13333	13354	13357	13364
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.54	0.54	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.56	0.56	0.56	0.56	0.56
77°F (25.0°C)	TC	10917	11442	11890	11970	12504	11573	11610	11903	11983	12517	12097	12111	12142	12156	12527	13066	13105	13136	13139	13151
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.58	0.58	0.59	0.59	0.59	0.58	0.58	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59
82°F (27.8°C)	TC	11042	11606	12074	12156	12712	11701	11746	12084	12169	12726	12245	12261	12299	12314	12737	13240	13290	13324	13328	13342
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68
86°F (30.0°C)	TC	11525	12101	12595	12683	13274	12154	12210	12609	12698	13288	12730	12745	12792	12810	13299	13782	13835	13868	13871	13884
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.82	0.82	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.84	0.84	0.84	0.84	0.84
95°F (35.0°C)	TC	12530	12598	13005	13082	13599	13317	13325	13329	13345	13616	13889	13891	13900	13901	13924	14966	14960	14963	14967	14987
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78						

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Heating Performance - Indoor Dry Bulb Temperature at 353 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	5128	5025	5000	4980	4935	0.88	0.89	0.89	0.90	0.90
-4°F(-20.0°C)	6974	6834	6800	6773	6712	0.95	0.96	0.96	0.96	0.97
0°F(-17.8°C)	7794	7638	7600	7570	7501	0.98	0.98	0.99	0.99	0.99
5°F(-15.0°C)	8820	8643	8600	8566	8489	1.01	1.01	1.02	1.02	1.02
14°F(-10.0°C)	9896	9698	9650	9611	9525	1.09	1.10	1.10	1.10	1.11
17°F(-8.3°C)	10255	10050	10000	9960	9870	1.12	1.13	1.13	1.13	1.14
22°F(-5.6°C)	10374	10167	10116	10075	9985	1.11	1.12	1.13	1.13	1.13
27°F(-2.8°C)	10493	10283	10231	10191	10099	1.11	1.12	1.12	1.12	1.13
32°F(0.0°C)	10611	10399	10347	10306	10213	1.10	1.11	1.11	1.12	1.12
35°F(1.7°C)	10683	10469	10417	10375	10282	1.10	1.11	1.11	1.11	1.12
37°F(2.8°C)	11534	11304	11247	11202	11101	1.13	1.14	1.14	1.15	1.15
42°F(5.6°C)	13664	13391	13324	13270	13151	1.21	1.22	1.23	1.23	1.23
44.6°F(7.0°C)	14516	14225	14154	14098	13971	1.25	1.26	1.26	1.26	1.27
47°F(8.3°C)	15793	15477	15400	15338	15200	1.30	1.31	1.31	1.31	1.32
52°F(11.1°C)	15313	15007	14932	14872	14738	1.18	1.19	1.19	1.19	1.20
57°F(13.9°C)	14833	14536	14464	14406	14276	1.06	1.07	1.07	1.08	1.08
59°F(15.0°C)	14641	14348	14276	14219	14091	1.02	1.02	1.03	1.03	1.03
62°F(16.7°C)	14353	14066	13996	13940	13814	0.95	0.95	0.96	0.96	0.96
64.4°F(18.0°C)	14122	13840	13771	13716	13592	0.89	0.90	0.90	0.90	0.90
69.8°F(21.0°C)	13604	13332	13265	13212	13093	0.76	0.77	0.77	0.77	0.78
75.2°F(24.0°C)	13085	12824	12760	12709	12594	0.64	0.64	0.64	0.65	0.65

Bolded values were tested. All other values are simulated and for reference only.